X(3872) Analysis Update

From ψ(2S) reweighting to the current PbPb strategy

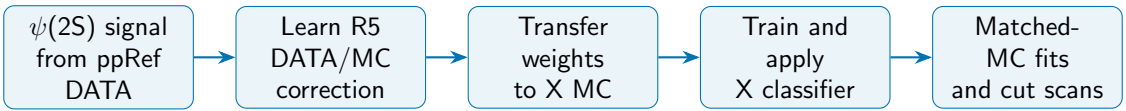
Lekai Yao

CMS

2026.7.29

ppRef prototype completed — PbPb selection and fit diagnostics in progress

Analysis path: validate the chain, then move to PbPb



ppRef question

Can the complete correction $\rightarrow$ ML $\rightarrow$ fit workflow be validated?

Interpretation boundary

Observed significance alone does not establish the final training preference.

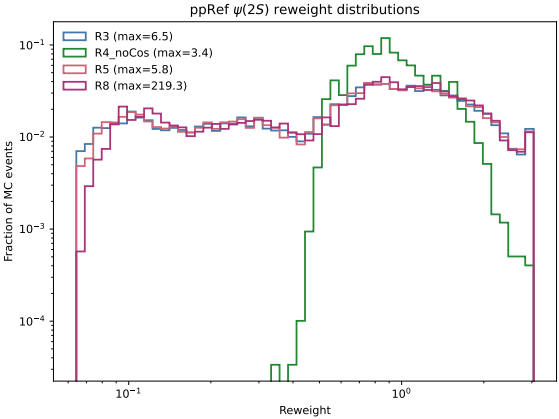
The ppRef prototype is now closed; the next bottleneck is PbPb ψ(2S).

ppRef ψ(2S) reweighting: R5 nominal

R5 inputs

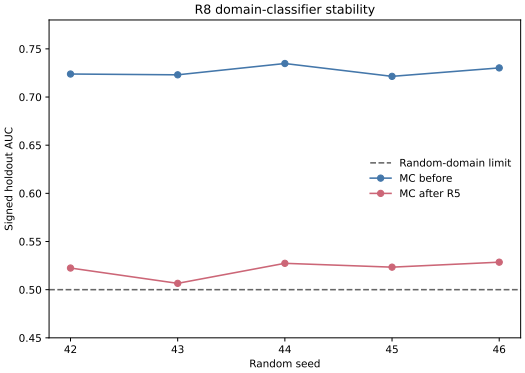
- `Bcos_dtheta`
- `Btktkpt`
- `Bchi2Prob`
- `Btrk1Pt`, `Btrk2Pt`

- sWeighted ψ(2S) DATA target
- Folded train/apply workflow
- R3/R4_noCos retained as checks
- R8 excluded after comparison



Source: `docs/ml_results.md`; R5 is the nominal ppRef prototype.

Closure: little residual DATA/MC separation



Independent holdout test

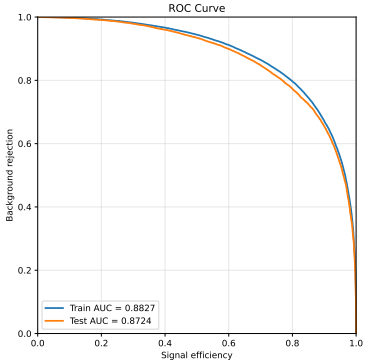
Domain-classifier AUC $\simeq 0.5225$, stable over five random splits.

- Weight statistics validated
- X MC support/coverage checked
- `Reweight` written to X MC
- No obvious residual separation

R5 passes the ppRef closure and transfer checks used in this prototype.

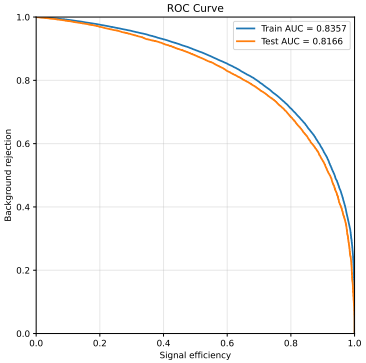
X classifier: unweighted and R5-weighted 8v2

Unweighted X MC



`X_pp24_v4_fid3_8v2_rw0_xgb_v1`

R5-weighted X MC



`X_pp24_v4_fid3_8v2_rwpsi2sr5v1_xgb_v1`

Both train/apply/draw chains and scored DATA/MC ROOT files are complete.

Matched-MC working points

Training	Bkg. eff.	Status	Yield	Signif.	Mass [GeV]
rw0	10%	0/0	762 ± 63	12.18	3.87028
rw0	3%	$-1/3$	286 ± 80	8.04	3.86884
rw0	1%	0/0	112 ± 20	5.52	3.86703
R5	10%	0/0	775 ± 65	12.21	3.87174
R5	3%	0/0	216 ± 54	5.84	3.87229
R5	1%	0/0	78 ± 23	3.40	3.87208

Interface fixed

Each training uses its own scored DATA and matched MC template.

Comparison caveat

rw0 3% did not converge; it cannot support a quantitative ranking.

No consistent R5 sensitivity gain across the three predefined working points.

Punzi optimum: useful diagnostic, not a final ranking

Training	Frozen cut	Efficiency	Fit status	Significance
rw0	0.875	0.2840	strategy-2: 0/0	7.01
R5	0.613	0.6775	0/0	16.62

What was validated

- Frozen before inspecting new fits
- R5 fit converges with a clear X peak
- R5 cut agrees with the later 0.6 scan point

Why this is not the answer

- Efficiency definitions differ
- rw0 cut 0.875 is too tight in DATA
- Significance cannot rank the schemes alone

Fixed-cut scans expose the relevant trend

Scan	Prediction range	Significance trend	Fitted mass range [GeV]
rw0	0.2–0.8	$20.27 \to 9.49$	$3.87128 \to 3.86914$
R5	0.2–0.8	$\sim 20 \to 6.63$	3.87113–3.87204

rw0

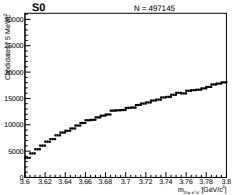
- Seven default fits converge
- Yield/significance fall with tighter cuts
- Frozen 0.875 is beyond the useful scan

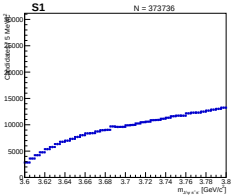
R5

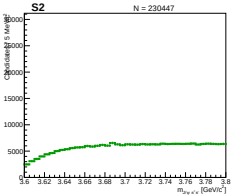
- More stable fitted peak position
- Cut 0.8 passes strategy-2 check
- $S(0.613) = 16.62$, $S(0.6) = 16.91$

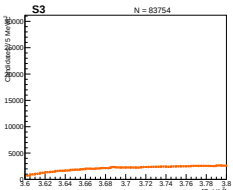
ppRef validates the chain, but does not uniquely choose the final correction scheme.

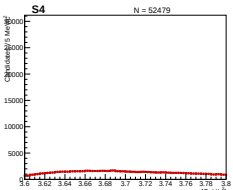
PbPb ψ(2S): S0–S4 selection scan completed











PbPb24 DATA

ψ(2S) selection pre-scan

$3.6 < m_{\mu^+\mu^-} < 3.8$ GeV/c^2

5 MeV/c^2 per bin

Raw counts; common y scale

No fit / no significance

H003 accepted

- 25.5M flat DATA entries
- S0–S4 spectra inspected
- S2/S3 selected for fits

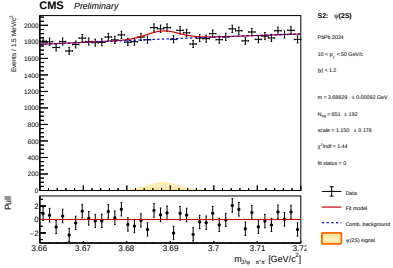
S0 baseline:

$$Q_B < 0.15,$$
$$|y_B| < 1.6,$$
$$p_{\mathrm{T}}^B > 10 \text{ GeV}.$$

S1–S4 and H004 use $|y_B| < 1.2$.

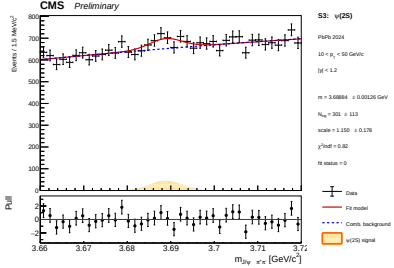
Current checkpoint: preliminary PbPb S2/S3 fits

S2 — preferred candidate



$$N_{\psi(2S)} = 651 \pm 192, \qquad \frac{S}{\sqrt{S+B}} = 4.39$$

S3 — tighter fallback



$$N_{\psi(2S)} = 301 \pm 113, \qquad \frac{S}{\sqrt{S+B}} = 3.41$$

H004 in progress

Both fits have status/covQual $= 0/3$; the DATA width scale reaches 1.15. Results are preliminary; sWeight and N_{eff} checks are pending.

Status and immediate next steps

Completed: ppRef prototype

- ψ(2S) signal target and R5 correction
- Closure and transfer to X MC
- rw0/R5 train, apply, matched fits
- Prediction scans and physical review

Next: PbPb-specific chain

1. Accept S2 or S3 fit configuration
2. Produce and validate sWeights
3. Train PbPb-specific correction
4. Retrain X ML and scan Prediction

The ppRef chain is validated.

The immediate decision is whether the preliminary PbPb ψ(2S) fit is robust enough to provide the next reweighting target.